Saudi Mais Co, for Medical Products Work Instruction	Doc. # : WI-8-2-6-06
	Rev. # : 0
Test Procedure for falling Drop Positive-Pressure Liquid Leakage of Small-Bore Connector	Date.# : 12.12.2022
	Page.# : 1 / 2

1. **Purpose:**

   To lay down the procedure for falling drop positive-pressure liquid leakage test of male and female small-bore non locking (slip) connector, locking connector with fixed threads & floating or rotatable collar Connectors.

2. **Scope:**

   This procedure is applicable for all small-bore connectors which are produced by Saudi Mais company.

3. **Definition:**

   A connector is assembled to a reference connector. Water is introduced into the connection and pressurized for the hold period.

4. **Responsibilities:**

4.1 Quality Controller are responsible to collect the sample as per AQL and test the connectors as per procedure during process and record the obtained results in the specified form.

4.2 Quality Control Manager is responsible to review the test and the test results as per standard requirements.

4.3 Quality Assurance Manager is responsible to verify and approve the test report.

5. **Equipment:**

5.1 Connector Assembly Device (**LAB-SBC-TMC-01**).

5.2 Pressure and Positive Pressure Liquid Leakage Tester (**LAB-SBC-TMC-02**).

6. **Environmental Test Conditions:**

   Perform test at a temperature within the range of 15˚C to 30 ˚C and a relative humidity between 25 % and 65 %.

7. **Procedure:**

7.1 Assemble the connector under test to the appropriate male or female reference connector both connectors being dry.

7.2 For a non-locking (slip) connector, assemble by applying an axial force of between 26.5 N and 27.5 N for 5 sec to 6 sec while rotating the connector under test to a torque of between 0.008 Nm. and 0.10 Nm. and a rotation not exceeding 90˚.

Prepared by : Q.C.M Javeed Gafoor	Approved by: Q.A.M Mr. Salman Rashid

Controlled Copy

7.3 For a locking connector with fixed threads, assemble by applying an axial force of between 26.5N and 27.5 N for 5 sec to 6 sec while rotating the connector under test to a torque of between 0.08 Nm. and 0.12 Nm.

7.4 For a connector with a floating or rot able collar, assemble by introducing the mating features (i.e., Taper Connector) together with an axial force of between 26.5N and 27.5N for 5 sec to 6 sec while rotating collar of the connector under test to a torque of between 0.08 Nm. and 0.12 Nm.

7.5 Ensure air supply pressure in at range of 0.4-0.6 Mpa.

7.6 Switch "ON" the pressure and positive-pressure liquid leakage tester device.

7.7 Click the touch screen "Pressure Liquid" button.

7.8 Set the test pressure and time.

7.9 Close 3 valves of the liquid tank and connect the test sample to valve 2 of the liquid tanks.

7.10 Connect the "AIR OUTLET" pipe of the device to the valve 3 of the liquid tank.

7.11 Open valves 1 and 2 and fill the funnel with RO water until water flows out of the sample end.

7.12 Close valve 1. Block the outlet of the sample, to prevent leakage.

7.13 Open Valve 3.

7.14 Touch "START" to start the test.

7.15 Maintain the pressure for the hold period specified in the relevant Application part of ISO 80369 while maintaining the assembled Connectors in the horizontal orientation.

7.16 Visually inspect for a falling drop of water from the Connection during the specified hold period.

8. Reference:

8.1 ISO 80369-20:2015 (ANNEX-C)

8.2 F-8-2-6-06

9. Modification / Documentation Change History:

Date	Changes	Revision
12-12-2022	Procedure Initiated	0

Prepared by : Q.C.M Javeed Gafoor	Approved by: Q.A.M Mr. Salman Rashid

Controlled Copy